

EDUCATION

International School, Vietnam National University, Hanoi (VNUIS)

BEng. Automation and Informatics

Hanoi, Vietnam

08/2021 - 07/2026

- QS WUR Ranking 386 for Engineering and Technology (Source)
- GPA: 3.71/4.0 (Rank #3 of the Cohort)
- Transcripts: [Click here](#)

National University of Singapore (NUS)

Exchange Semester I, AY25/26

Singapore

08/2025 - 12/2025

- QS WUR Ranking 12 for Engineering and Technology (Source)
- Courseworks: Robot Mechanics and Control (ME4245), Electric Drives and Control (EE4502), Calculus (MA2002).

RESEARCH EXPERIENCE

AIM-HI Institute, National Chung Cheng University

Research Intern

Chiayi, Taiwan

05/2025 - 07/2025

- **Project:** Smart Sensing Technology for Smart Manufacturing.
 - * Developed a data-driven approach to estimate dynamic cutting force model in milling process by combining nonlinear system analysis with qualitative simulation.
 - * Processed and trained 12 datasets of the cutting forces, each containing multi-channel sensor signals (three-axis dynamometer, spindle speed, feed rate, and depth of cut).
 - * Implemented visualization and error analysis of predicted vs. actual measured cutting forces using Python.
 - * Results: Prediction accuracy of 60-70% compared to experimental ground-truth cutting force measurements.
 - * **Supervisor:** Prof. Her-Terng Yau

Remote Research

08/2025-Current

- **Project:** Prediction of cutting force model using AI and data-driven methods
 - * Implementing 03 machine learning models and 01 data-driven model to predict cutting force.
 - * Training 27 experienced datasets collected by Taguchi method.
 - * **Supervisor:** Prof. Her-Terng Yau

Intelligent Control and Artificial Intelligence Lab, VNUIS

Student Research

Hanoi, Vietnam

09/2022 - Current

- **Project:** Implementation Model Predictive Control for Machining System (Current)
 - * Reviewing more than 30 papers in control of machining system and cutting force prediction.
 - * Combining machine learning and MPC model to control machining system.
 - * **Supervisor:** Assoc. Prof. Nguyen Nhu Tung
- **Project:** Implementation Adaptive Control for 3-DOF Parallel Robot (2024)
 - * Designed mechanical model and physical prototype of 3-DOF delta parallel robot.
 - * Applied adaptive fuzzy logic combined with backstepping control for robust trajectory tracking under uncertainties.
 - * Simulation Results (Matlab/Simulink):
 - Reduced steady-state deviation at J2 from 1.2 rad (Backstepping) to nearly 0 rad (Backstepping + Fuzzy).
 - Settling time improved from 0.3 s to 0.15–0.2 s.
 - Coordinate tracking (x, y, z) stabilized with maximum deviation 0.35 rad.
 - * Results: Published in Lecture Notes in Networks and Systems, Springer, Q4.
 - * **Supervisor:** Dr. Le Xuan Hai
- **Project:** Detection of Personal Protective Equipment (PPE) in Construction Sites (2023)
 - * Built a custom dataset of 2,632 real construction site images across 5 PPE classes (helmet, vest, gloves, boots, person)
 - * Trained and evaluated YOLOv5s with 6 activation functions, SiLU achieved the best performance.
 - * Results: Precision 84.8%, Recall 80.2%, mAP@0.5 83.9%, mAP@0.5–0.95 43.6%.
 - * **Supervisor:** Dr. Kim Dinh Thai

TEACHING EXPERIENCE

Faculty of Engineering and Technology, VNUIS

Teaching Assistant

Hanoi, Vietnam

09/2023 - 12/2024

- Modeling and Simulation of Control Systems (Semester 1, AY24/25)
 - * Assisted in teaching undergraduate course focused on modeling, analysis, and simulation of dynamic systems.
 - * Supported lecturer in preparing tutorials, lab manuals, and course materials.
 - * Mentored student groups on applying simulation tools for robotics, mechatronics and process control applications by using MATLAB/Simulink.
 - * **Supervisor:** Dr. Le Xuan Hai
- PLC Programming (Semester 1, AY24/25)
 - * Assisted in preparing technical manual documents in Siemens S7-1200 and TIA Portal.
 - * Mentored 01 team programming 4-floor elevator control logic project and 01 team programming pneumatic valve and signal control project.
 - * Assisted with troubleshooting, hardware interfacing, and simulation.
 - * **Supervisor:** MSc. Pham Hai Yen
- Microprocessors and Microcontrollers (Semester 1, AY23/24)
 - * Assisted in teaching undergraduate course covering embedded C, and microcontroller interfacing (Arduino, STM32).
 - * Mentored students on debugging techniques using Keil uVision, Arduino IDE, and STM32CubeIDE.
 - * Supported lecturer in preparing tutorials, lab manuals, and course materials.
 - * **Supervisor:** Dr. Le Xuan Hai

Hung STEM Robotics Lab

Collaborator

Hanoi, Vietnam

07/2022 - 12/2022

- **Project:** World Robotics Olympiad 2022 Vietnam
 - * Mentored in a team to participate in World Robotics Olympiad 2022 Vietnam, assisted the team in building up new LEGO modules.
 - * Engaged in teaching STEM robotics classes using LEGO NXT in STEM club at Grammar Newton International Bilingual School, Hanoi.
 - * **Supervisor:** Mr. Nguyen Quoc Hung (Email)

WORKING EXPERIENCE

ETEK Automation Solutions JSC

Intern

Hanoi, Vietnam

03/2025 - 05/2025

- **Project:** AGV-based Virtual Simulation for Automated Warehouse Logistics – VINFAST Hai Phong.
 - * Sketched AGV/AMR traffic intersections using AutoCAD and simulated their operations using Visual Components Premium 4.6.
 - * Monitored and analyzed AGV/AMR behavior, identified errors, and troubleshooting.
 - * **Supervisor:** Mr. Nguyen Ngoc Khanh (Email)

Vietnam E-Commerce and Digital Economy Agency, Ministry of Industry and Trade

Collaborator

Hanoi, Vietnam

07/2023 - 09/2023

- **Project:** Program for Consultancy Support and Sales Policy Development for E-Commerce Website (CPAPER)
 - * Coordinated with SMEs across Dak Nong and An Giang province to develop e-commerce platforms.
 - * Assisted in website design, content management, and prepared technical manual documents.

PUBLICATION

1. **Long Nguyen Khac.** et al. *Adaptive Fuzzy Logic in Backstepping Control for 3-DOF Parallel Robot.* International Conference on Advances in Information and Communication Technology, doi: 10.1007/978-3-031-80943-9_57.
 - Main highlights: an adaptive fuzzy logic control system integrated with backstepping control for a 3-DOF delta robot.
2. **Long Nguyen Khac.** et al. *YOLOv5s Model with Applied Activation Functions for Personal Protective Equipment Detection in Construction Sites.* Proceedings of National Conference on Smart Technology Applications in Industry 4.0 Smart City, and Sustainability. (2023)
 - Main highlights: a deep learning-based approach using YOLOv5s with 06 activation functions with a new dataset of 2632 images for detecting PPE.

TALKS

The 3rd International Conference ICTA 2024

Phu Tho, Vietnam

Mechatronics and Automation

11/2024

HONORS AND AWARDS

- Selected as one of 20 VNU students received **DiscoverNUS & NUS ASEAN Master's Scholarship** for one semester exchange fully funded by National University of Singapore. (Semester 1, AY25/26)
- Selected as one of 28 global students received **Taiwan Experience Education Program (TEEP) Scholarship** fully-funded by Ministry of Education, Republic of China (Taiwan) to intern at National Chung Cheng University (CCU).
- Selected as one of 04 VNU students to participate in **ASEAN in Today's World** in Indonesia funded by Kyushu Univeristy, Japan.(2025)
- Selected as one of 20 undergraduate and graduate students received **TOSHIBA scholarship**. (AY 24/25)
- Received **Academic Encouragement Scholarship**, 4/6 semesters.(Semester 1, AY22/23, Semester 1, AY 23/24, Semester 1,2 AY 24/25)
- Selected as one of 20 VNU students to participate in **Temasek Foundation Specialist's Community Action & Leadership Exchange Program** in Singapore fully funded by Temasek Foundation. (2024)
- Received **Outstanding Young Face Award**, VNUIS. (2024)
- Received **Student with Five Good Merits Award**, VNU. (2024)
- Received **Fighting Spirit Prize** in South-East Asia Circuit and System Society Hackathon. (2022)

SKILLS

General skills: Mathematical modeling, control systems designing, embedded programming, mechanical designing

Programming Skills: Matlab, Python, Embedded C/C++

Specialisation: Adaptive Control, Machine Learning, Robotics

Softwares and Frameworks: Matlab/Simulink, Solidworks, PyTorch

Languages: Vietnamese (Native), English (IELTS 6.5: Writing 7.0), Chinese (Beginner)

REFERENCES

- **Prof. Her-Terng Yau**
Director of AIM-HI, National Chung Cheng University, Taiwan
Email: yauherterng@gmail.com
- **Assoc. Prof. Nguyen Nhu Tung**
Senior Lecturer, Faculty of Engineering and Technology
Academic Advisor of Industrial Systems Engineering and Logistics Major
Email: tungnn@vnu.edu.vn
- **Dr. Le Xuan Hai**
Academic Advisor of Automation and Informatics Major
Director, Intelligent Control and Artificial Intelligence Lab (ICL)
Vice Dean, Faculty of Engineering and Technology, International School, VNU
Email: hailx@vnu.edu.vn